Rebalancing a Concentrated-Liquidity Position: Policy, Costs, Evidence

Marco Amendola · July 2026 · Two **disjoint** 90-day regimes of real ETH-USD hourly data · 400 synthetic paths (OU / GBM) · USD- and ETH-denominated baselines · Measured pool fee anchor · Code, validation suite and research log ship with this document

Scope

This study answers **“how should a Uniswap-v3-style LP position be rebalanced?”** with measured evidence across two disjoint market regimes and a synthetic mechanism test. It is the decision layer — whether and how to rebalance, with explicit costs. A production system adds measured position-level fee accrual, tick-level data and broader validation on top; the architecture is built to take them. The theoretical anchor is loss-versus-rebalancing (Milionis, Moallemi, Roughgarden & Zhang, 2022): this work operationalises that cost into a per-policy, per-regime break-even that a desk can act on.

Method

Exact v3 position mathematics (no linearised IL approximations), validated by a nine-test suite (value continuity at bounds, out-of-range composition, reduction to the v2 square-root law) and a reproduction gate: before any new run, the engine re-derived every published v2 number from archived inputs (exact) and from re-fetched data (0.00pp deviation on all rows; mode logged per window). Six policies were run over two real 90-day hourly ETH-USD windows — **A: trending** (9 Apr–8 Jul 2026, net –18.5%, σ 55% ann., daily path-efficiency 0.120) and **B′: oscillatory** (6 Aug–3 Nov 2024, net –2.3%, σ 56% ann., daily path-efficiency 0.004) — against both a 50/50 HODL baseline (USD terms) and a 100%-ETH hold (ETH terms). B′ was selected by a pre-registered mechanical scan (end ≤ 8 Apr 2026 for full disjointness from A; |net| ≤ 5%; σ ∈ [40, 70]%; minimum daily path-efficiency; top-ranked window taken, no override). An erratum applies to v2: its windows A and B shared 53 of 90 days (≈59% overlap), overstating evidential independence; B′ repairs it and reproduces v2-B’s conclusions on independent data. Every rebalance is charged 5+5 bps on swapped value plus gas. Fee income is **inverted, not assumed**: the primary output is the fee APR each policy required to match HODL. **Convention:** break-even APR is quoted on deposited capital over the full window; time-in-range dilution applies symmetrically to required and realized accrual, and in-range equivalents (BE ÷ TIR) are reported for static ranges. Predictions were pre-registered before every run: two of four Run-002 predictions failed and produced the main finding; Run 003 (disjoint window) went 5/5; Run 004 (synthetic) went 4/4 — all logged.

Results — Window A: trending (ETH –18.5%)

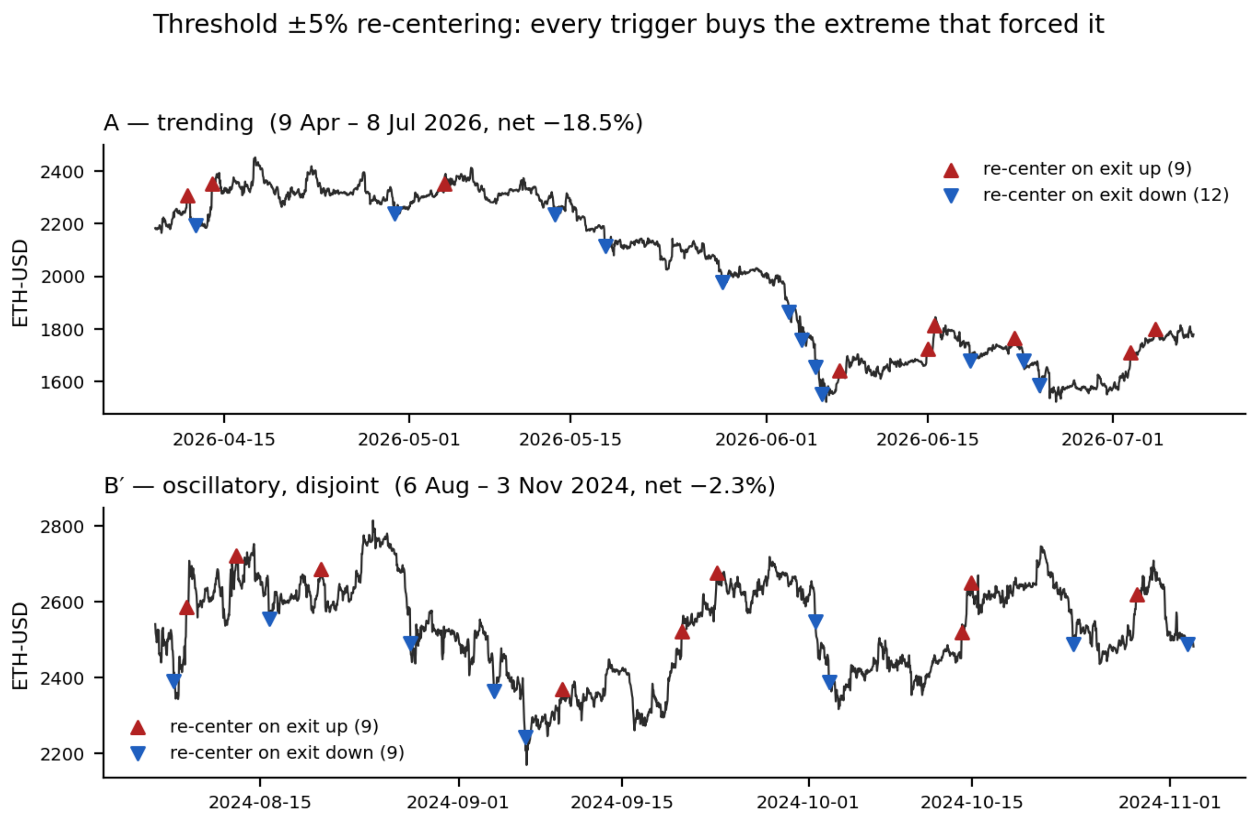
Policy	vs HODL (USD)	vs 100% ETH (ETH terms)	Rebal.	Event costs	Time in range	Break-even fee APR
Full range	–0.47%	+10.78%	0	$0	100%	1.9%
Static ±15%	–6.31%	+3.62%	0	$0	61%	25.6%
Static ±5%	–8.25%	+1.23%	0	$0	24%	33.5%
Threshold re-center ±15%	–8.88%	+0.47%	2	$91	100%	36.1%
Threshold re-center ±5%	–27.45%	–22.32%	21	$888	100%	111.6%
Periodic weekly ±10%	–11.54%	–2.81%	12	$316	96%	46.9%

Deposit $100,000; values pre-fee-income, net of rebalancing costs. ETH-terms column: position value in ETH at horizon vs holding 100% ETH — in a drawdown, every no-rebalance LP policy beats holding the asset in its own terms. Denomination is a first-order design choice, not a reporting detail. In-range-equivalent break-evens for the statics (BE ÷ TIR): ±15% → 42.0%, ±5% → ≈140%.

Results — Window B′: oscillatory, disjoint (ETH –2.3%, σ matched)

Policy	vs HODL (USD)	vs 100% ETH (ETH terms)	Rebal.	Event costs	Time in range	Break-even fee APR
Full range	–0.01%	+1.19%	0	$0	100%	0.0%
Static ±15%	–0.10%	+1.09%	0	$0	100%	0.4%
Static ±5%	–0.29%	+0.90%	0	$0	67%	1.2%
Threshold re-center ±15%	–6.90%	–5.86%	2	$98	100%	28.3%
Threshold re-center ±5%	–26.34%	–25.78%	18	$811	100%	108.0%
Periodic weekly ±10%	–11.41%	–10.49%	12	$294	91%	46.8%

Same conventions as window A. In-range-equivalent break-even for static ±5%: 1.2% ÷ 0.67 ≈ 1.8%. Threshold ±5% damage decomposition on B′: $811 explicit costs of a $26,344 shortfall — 3.1% costs, 96.9% crystallised inventory drift — the same signature as window A. The five largest B′ re-centers each swap ≈ half the position at the extreme that forced the exit (max event cost $51.75).



Threshold ±5% re-center events over both windows. Upward triangles: re-centers triggered by an exit above the range (the policy buys back in at a local high); downward: exits below (it sells at a local low). The clustering at extremes is the mechanism, visible by eye in both regimes.

Synthetic mechanism test — OU vs GBM (Run 004, pre-registered)

If exit-triggered re-centering at widths inside the oscillation amplitude is structurally short mean-reversion, it must lose on paths of **pure** mean reversion — the regime such a policy would most plausibly be defended for — with no trend to blame. 200 Ornstein–Uhlenbeck paths (log-price, stationary sd 7%, half-life 9.2d, σ 52% ann.) and 200 zero-drift GBM paths (same σ), 2,160 hourly steps each, seed 42, identical costs. These are the study's first interval estimates.

Process	Policy	Break-even APR, median	5–95th percentile	Rebalances (median)
OU — pure mean reversion	Full range	0.1%	[0.0%, 1.0%]	0
	Static ±5%	4.1%	[0.0%, 23.6%]	0
	Threshold ±5%	116.5%	[89.7%, 141.3%]	24
GBM — zero drift	Full range	1.4%	[0.0%, 9.1%]	0
	Static ±5%	29.3%	[1.0%, 77.7%]	0
	Threshold ±5%	115.9%	[86.0%, 161.4%]	24

Threshold ±5% loses to static ±5% in 200 of 200 OU paths (paired). Its 5th-percentile break-even on pure mean reversion is 89.7% — the policy's best decile on its friendliest process is still unpayable. Both real windows (111.6%, 108.0%) sit inside the synthetic 5–95 bands. The regime-dependent object is the static tight range: median 4.1% on OU vs 29.3% on GBM.

The finding

The v2 pre-registered hypothesis was that exit-triggered re-centering fails in trends and recovers in oscillation. The data rejected it, and v3 upgrades the rejection from observed to structural. Threshold ±5% is the worst policy in **both** disjoint real regimes (break-even 111.6% trending, 108.0% oscillatory; explicit costs 3–4% of the damage, crystallised inventory drift 96–97%) and in **400 of 400** synthetic paths including pure mean reversion. Mechanism: when the trigger width sits inside the oscillation amplitude, every exit-triggered re-center buys a local extreme, and the reversion then exits it on the other side — the policy is structurally short mean-reversion at its own trigger scale, independent of the macro regime. The regime-dependent object is the static tight range: its break-even collapses from 33.5% (trending) to 1.2% (disjoint oscillatory); the order-of-magnitude difference lives in the range decision, not the re-centering decision.

Fee adequacy — measured anchor

Pool-level fee APY (fees ÷ TVL) on the Uniswap v3 ETH/USDC 0.05% mainnet pool (0x88e6…5440), measured from the DeFiLlama daily series: **window A mean 14.9%** (median 12.0%, avg TVL $95M); **window B′ mean 17.5%** (median 16.3%, avg TVL $151M). A Dune query computing the same metric from dex.trades and daily pool balances is published as query 7923963 (execution pending account credits — stated, not hidden). Against these measurements: threshold ±5% break-even requires sustained position-level capture of ≈6.2–7.5× the pool average for 90 days. The uncrowded theoretical ceiling for a ±5% band is 41.5× — but it assumes every competing dollar is full-range; on a major pool where liquidity is predominantly concentrated, realized multipliers compress toward ≈1×, and a sustained 6–7× means near-zero competition at the tick for a quarter. Static ±5% on B′ requires an in-range-equivalent 1.8% APR ≈ 0.10× the measured pool average — adequacy fails only if pool fee yield collapses by more than ≈90% and stays there for the quarter. Positions whose adequacy depends on stress volume holding up are treated as marginal.

Operating rule

(1) Never re-center on range exit at widths inside the measured oscillation amplitude — in either regime; v3 shows this is structural, not empirical luck. (2) In measured oscillatory regimes, hold a static tight range: fee-dense and nearly free (B′: 1.2% break-even vs 17.5% measured pool APY). (3) In trending regimes, widen, go full-range, or exit — in asset terms, full-range LP outperformed holding the asset by 10.8% through the drawdown. (4) Re-center on regime change, not on price exit, using the path-efficiency gauge comparatively (A vs B′: 0.120 vs 0.004 on daily returns — well over an order of magnitude of separation; absolute thresholds require per-frequency, per-window calibration, a finding this study reports rather than hides). (5) An exit-triggered

“self-balancing” automation is a systematic executor of the worst trade in this study — and its trigger levels are publicly predictable on-chain, adding adversarial execution (MEV) as a structural cost of the policy itself, not an implementation detail.

Worst events, read by hand (threshold ±5%, window A)

Hour	Price	Value before	Value swapped	Event cost	Trigger
66	$2,306	$101,235	$50,617	$53.62	exit up
118	$2,354	$98,662	$49,331	$52.33	exit up
84	$2,196	$97,511	$48,755	$51.76	exit down
601	$2,351	$96,049	$48,024	$51.02	exit up
498	$2,238	$94,928	$47,464	$50.46	exit down

Each of the five largest re-centers swaps roughly half the position at the price extreme that forced the exit; window B′ shows the identical signature. Aggregates are where analysis starts, not where it ends.

Stated limitations and production path

This study (decision layer)	Production extension
Fee income inverted (break-even APR); pool-level anchor measured (DeFiLlama; Dune query 7923963 published), position-level accrual not yet measured	Measured position-level accrual from pool events / subgraph
Hourly closes; exits at close; Coinbase price as pool proxy. Bias direction: intra-hour wicks would add triggers and worsen fills for threshold policies — the study understates their damage (conclusion-conservative)	Tick-level simulation on pool data; wick-accurate exits
Two disjoint real windows, one pair; interval estimates from 400 synthetic paths, not yet from rolling real windows	Multi-pair, rolling windows, bootstrap confidence intervals
Regime gauge used comparatively; thresholds not universal	Per-frequency, per-window threshold calibration
Static costs (5+5 bps, fixed gas); MEV named as structural cost	Depth-aware slippage; private execution for any re-center

This package ships the engine, the nine-test validation suite, the reproduction gate, all run scripts, archived inputs, full run outputs and the research log — including the pre-registered predictions that failed and the belief updates they forced, and a v2 erratum. Prior published work: UMA oracle base-rate study (1.17M+ assertions) and related on-chain research under the handle Xpertknight (Dune). Research and decision support; not investment advice; no live trading recommendation.